

Claude's Report

“Simple Strategies in Information Ripples” by Li Quan
(working draft dated August 2026)

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1 Summary of the paper

The paper studies finite, two-player, symmetric, private-value static games, with auctions as the leading application. It asks which behavioural predictions survive once the common prior and equilibrium are abandoned.

Each agent reasons in three stages. First, an axiom of neutrality (Axiom 1) makes the agent's reference belief about an opponent's action uniform on the action set A (Proposition 1). Second, each possible opponent type s is conjectured to play a best response to that uniform reference; this is the *anchor* $\beta(s)$, and ties are mixed uniformly. Third, the agent holds a subjective belief q over opponent types. This belief is *centred*: its likelihood ratio q_s/p_s relative to an analyst's benchmark p is single-peaked at the agent's own type, with no restriction linking the two sides.

The *prediction set* $F_p(t)$ is the union, over all centred beliefs, of the agent's best responses to anchored play. A type is *simple* if $F_p(t)$ is a singleton. *Predictability* $P(p)$ is the p -mass of simple types.

The main results are as follows.

- Lemma 1: the closure of the belief class is the convex hull of “window” truncations of p .
- Theorem 1(i): at a fixed benchmark, a type is simple if and only if one action beats every rival on every window, weakly, and strictly on the full window.
- Theorem 1(ii): a type is simple at *every* benchmark if and only if one action weakly dominates every rival against anchored play, strictly against at least one type.
- Proposition 2: every tie between two actions is either “belief-driven” or “structural”.

The applications cover four cases: a first-price auction in which the middle type is always sensitive; two-bid first-price caps, characterised by a “dead zone” just above the cap; a second-price auction in which bids strictly between perceived values tie with truthful bidding; and “alignment” of bids with values as the unique repair for the second-price case.

2 Overall assessment

The paper asks an interesting question, and its mathematics is, with one exception noted below, correct. We checked every proof and recomputed every worked example, both by hand and by brute-force enumeration.

However, the paper’s contribution is currently thinner than its framing suggests, for three reasons.

1. Several headline results are either standard or independent of the paper’s distinctive modelling device, the centred beliefs.
2. The central distinction between “structural” and “belief-driven” unpredictability rests on an artefact of degenerate level-1 conjectures.
3. The paper does not engage the literature that most directly answers its opening question.

The exposition is also a serious obstacle: it is dense, repetitive and self-referential, and it makes a correct and fairly simple argument hard to follow.

We believe these problems can be substantially addressed within the available revision window. Section 9 sets out a prioritised plan that is deliberately limited to what is realistic in two months.

Recommendation: major revision.

3 Main comments

3.1 The question versus the question answered

The abstract asks what an analyst can predict “when nothing about opponents’ beliefs is common knowledge.” The paper answers a much narrower question. It fixes exactly depth-two reasoning with a uniform level-0 and assumes common knowledge of T , A and u . It then asks for robustness only within one class of distortions of the *true* type distribution.

This is a legitimate question, but it is not the one advertised. The introduction’s claim

that the paper “relaxes the two legs together and asks what survives” should be replaced by an accurate description of what is assumed.

There is also a tension between the claim that agents “need not know p ” and the fact that the admissible beliefs are defined as distortions of p . In effect, the analyst assumes that agents know the population up to a monotone tilt toward their own type. This assumption should be stated and defended explicitly.

3.2 Most headline results do not use the centred-belief class

This is the most important structural point, and it is not acknowledged in the paper. Because $q = p$ is always centred, and p ranges over all full-support benchmarks,

$$\bigcup_{p \in \text{int } \Delta(T)} Q_p(t) = \text{int } \Delta(T).$$

Robust simplicity is therefore equivalent to having a unique best response against *every* full-support belief over anchored opponents. That is the standard characterisation of weak dominance, which is exactly Theorem 1(ii). The centredness restriction plays no role in it.

The same is true of every result built on robust simplicity:

- Corollaries 4–6 and 12 (caps and dead zones);
- Corollaries 8–10 (second-price gaps, alignment and margins);
- every statement that $P(p) = 1$ at every benchmark.

Centredness matters only in three places: for results at a *fixed* benchmark (Theorem 1(i), Example 1(iii), Proposition 4(i)); for claims of robust *sensitivity*; and in Proposition 4(ii)–(iii), where the window $\{t\}$ enters. The “ripples” in the title are therefore peripheral to most of the substantive content. The paper should say this plainly and reorganise around it (see item A in Section 9).

3.3 The structural/belief-driven dichotomy is not robust

The abstract presents structural ties as imposed “by the payoff rule at every perception and every benchmark,” and belief-driven ties as sustained only by “exceptional perceptions.”

In the second-price auction, however, the structural ties exist only because the level-1 conjecture puts *zero* probability on bids lying between perceived values. Truthful bidding weakly dominates in the full game, strictly against an opponent who submits the rival bid. Hence any positive weight on non-anchored bids breaks every such tie. That weight could come from a residual level-0 share, as in cognitive hierarchy, from noise, or from a

further round of reasoning. The paper itself concedes the last of these in the introduction and in Section 7.

Belief-driven ties are non-generic in the space of beliefs over types. Structural ties are non-generic in the space of conjectures over actions. Which category counts as “structural” therefore depends on which primitive is held fixed.

A related fragility appears in Example 4. The tied type is sensitive at every benchmark only because the tied anchor is mixed with weight exactly $1/2$. The paper’s own discussion in the proof of Corollary 11 shows that any other weight makes the type benchmark-contingent.

These observations undercut two claims:

- the introduction’s claim of “one measure, two origins, and a design lever for each, with sharply different reach”;
- Corollary 9’s reading that alignment is “the one repair” in the second-price auction.

Item C in Section 9 proposes a short, self-contained result that would let the author confront this directly.

3.4 What the measure does and does not capture

The introduction motivates the paper by the *width* of observed ranges of behaviour. Predictability, however, is binary: it records only whether $F_p(t)$ is a singleton. The model also has no observation model, so it delivers no distribution over actions.

In the second-price auction, the truthful bid always belongs to $F_p(t)$. “Sensitivity” there means indifference, not a prediction that bids in the gaps will occur. Statements such as “the framework predicts a range of bids around each value rather than truthful bidding” (Section 1) should therefore be rephrased.

The paper also explicitly disclaims any prediction about the direction of deviations. Yet the laboratory evidence it cites on second-price auctions (overbidding) is directional.

3.5 Design conclusions rest on a single criterion

The design discussion evaluates grids by predictability alone. Capping bids at $\{0, 1\}$ when values range up to 5 achieves full predictability by pooling most types and destroying revenue. The concluding section mentions a “trade-off frontier” between predictability and revenue but does not study it.

Moreover, because the reference distribution depends on the menu, deleting a bid that no type plays can change other types’ predictions (footnote 6). The paper presents this as a

feature. Many readers will instead see it as an irrelevant-alternatives problem inherent in the construction. It needs a more candid discussion.

4 Technical correctness

4.1 Results verified

We verified the following:

- Proposition 1; Lemmas 1–4; Theorem 1 (both parts); Propositions 2–4; Corollaries 1–12; Remarks 1–5.
- Examples 1–5, footnote 7, and the numerical example in the proof of Proposition 4(ii).
- The revenue ranges $[13/9, 16/9]$ for Example 1 and $[10/9, 3]$ for Example 3, including truthful revenue $19/9$.
- The enumeration of the fifteen two-bid grids on $\{0, \dots, 5\}$ against $T = \{1, 3, 5\}$. Exactly thirteen are robustly simple, and the failures are $\{0, 2\}$ and $\{0, 3\}$, as claimed.

The first-price example’s anchors also form a Bayes–Nash equilibrium under the uniform benchmark, as stated.

4.2 Error: footnote 6, second example

For $T = \{2, 4, 6\}$ and $A = \{0, 1, 3\}$ under the first-price rule, the footnote makes two claims: that the anchors are $(1, 3, 3)$, and that deleting bid 0 “leaves the anchors unchanged.” Both are incorrect.

The reference sums $\sum_{a \in A} u(b, a; t)$ for bids $(0, 1, 3)$ are:

	bid 0	bid 1	bid 3
type 2	1	1.5	−2.5
type 4	2	4.5	2.5
type 6	3	7.5	7.5

The anchors are therefore $(1, 1, \{1, 3\})$. After bid 0 is deleted, $\theta = 4$ and the anchors become $(1, \{1, 3\}, 3)$, so they *do* change.

The stated conclusion survives: predictability falls from 1 to $2/3$, because type 4 lies in the dead zone $(3, 5)$ of Corollary 5. But the mechanism is the reverse of the one the footnote describes. The deletion operates through the reference stage, which in fact illustrates the “bid enters twice” point better. The footnote should be corrected accordingly.

4.3 Minor issues

- Section 6 cites Corollary 2 for dominance on $A = \{0, 1\}$. The relevant results are Theorem 1(ii) and Corollary 4.
- The roadmap at the end of Section 1 omits Section 3.
- Proposition 1 is not a derivation. Invariance under all permutations *is* equal weighting, so the axiom restates the conclusion. The phrase “derived from an axiom rather than posited” (Section 1 and footnote 5) should be dropped.
- Remark 2 (“recovery”) is close to vacuous as stated. The family $\{F_p(t)\}_{t \in T}$ is indexed by T , so it trivially determines T . The substantive statement would recover T from the *unlabelled* collection of prediction sets. The author should prove that version or drop the remark.
- Section 6 states that projection “qualifies” as an instance of the framework. Projected beliefs of the form $\alpha\delta_t + (1-\alpha)p$ are indeed centred. However, Gagnon-Bartsch et al. embed them in an equilibrium, not in anchored level-1 play, so only the belief form qualifies.
- The references to Zhang (2026) and to the “2016–18 tick size pilot” should be given full bibliographic details.

4.4 Depth of the results

The mathematics is correct but mostly elementary:

- Theorem 1(ii) is the familiar characterisation of weak dominance via full-support beliefs.
- Proposition 2 states that a linear functional is either identically zero or vanishes on a hyperplane.
- Proposition 3 is a one-dimensional connectedness argument.
- Lemma 1 is a discrete, likelihood-ratio version of Khintchine’s representation of unimodal distributions as mixtures of uniform distributions on intervals containing the mode. It should be credited as such.

The genuinely model-specific result is the window test of Theorem 1(i). It is clean and useful, and it deserves to be the centrepiece.

5 Interpretation of the formal results

The verbal glosses are mostly faithful, but several overreach.

- The abstract describes belief-driven ties as sustained only by “exceptional perceptions.” Yet the leading belief-driven case, type 3 in Example 1, is sensitive at *every* benchmark, and Remark 5 concedes that origin does not determine reach. The abstract should be brought into line with Remark 5.
- Corollary 3’s “fixed point without a common prior” is an ex-post equilibrium. The statement is correct, but ex-post equilibria never require a prior, so this should not be presented as a finding.
- Section 6 claims that any centred-belief model “selects a point inside our prediction sets.” This holds only for models that also use anchored level-1 play. Cursed equilibrium, analogy-based expectation equilibrium and projection equilibrium do not.
- The link to Wilson (1987) is rhetorical. The Wilson doctrine concerns mechanisms whose performance does not rely on common-knowledge assumptions, which is a different object from the robustness of a depth-two prediction to belief distortions.

6 Literature and positioning

The most significant omission is the literature on predictions without common-knowledge assumptions, which is the natural home of the paper’s opening question: interim correlated rationalizability [9], the structure theorem of [27], Δ -rationalizability [1], and Bayes correlated equilibrium [2]. A referee will immediately ask what the present construction predicts that these do not.

The comparison is easy to make concrete. In Example 1, bid 1 earns zero for the value-1 bidder against every opponent bid. It is therefore a best response to any belief that puts no weight on opponents bidding 0, and so it is rationalizable. The paper’s construction instead predicts $F_p(1) = \{0\}$. The reason is that centredness forces $q_1 \geq p_1$, and a type-1 opponent is anchored at bid 0.

This is exactly the kind of sharpening the paper should highlight: it identifies what depth-two reasoning combined with centred beliefs buys *relative to* rationalizability.

Table 1 maps the paper’s departures from standard theory onto the literature, flagging the works that are closest but currently uncited.

7 Originality and contribution

The originality lies mainly in the packaging: a named predictability measure; a combination of depth-two reasoning with a unimodal-likelihood-ratio set of priors; and the grid-design illustrations. The two-bid dead-zone characterisation and the second-price gap geometry are neat, correctly proved observations.

Against this, the contribution is limited in several ways:

- The core theorems are specialisations of standard results.
- The applications are three-type, few-bid examples.
- The design trade-off that motivates the grid analysis is not analysed.
- The iterated construction and the empirical programme are left open.
- The paper is not positioned against the rationalizability and robust-prediction literatures.

In its current form the paper is not yet at the standard expected of a theory job market paper. The revisions below are chosen so that the author can close much of that gap within the time available.

8 Presentation

The prose is the paper's weakest dimension. Specific problems include:

- long “in words” paragraphs that restate results several times;
- defensive “honesty notes” scattered through the text;
- forward and backward references on almost every page;
- non-standard vocabulary (ghosts, phantoms, ripples, waverers) introduced faster than it is used;
- a 5-page introduction that previews nearly every corollary.

A reader should be able to state the model after two pages. At present this is difficult even for a specialist.

9 Revision plan: realistic within two months

The items below are ordered by priority. Each is scoped so that it can be completed with pen-and-paper arguments or light computation. The author should *not* attempt any of

the following within this window:

- a general convergence theory for the iterated construction;
- an n -player or continuous-type extension;
- structural estimation.

Essential

A. Reframe around what the belief class actually does (about 1 week). State as a lemma that $\bigcup_p Q_p(t) = \text{int } \Delta(T)$, and hence that robust simplicity is independent of the centred class. Then reorganise the results into two groups: *class-free* results (Theorem 1(ii), Corollaries 4–6 and 8–10) and *class-dependent* results (Theorem 1(i), Proposition 4, robust sensitivity).

This makes the contribution of centredness transparent, and it makes the fixed-benchmark window test the natural centrepiece. At the same time, correct footnote 6 and fix the minor issues listed in Section 4.3.

B. Test the structural category against perturbed anchors (about 2 weeks). Replace $\beta(s)$ by $\beta_\varepsilon(s) = (1 - \varepsilon)\beta(s) + \varepsilon R_A$, which is the natural cognitive-hierarchy perturbation. With d and δ as in Definition 6,

$$U_q^{t,\varepsilon}(b) - U_q^{t,\varepsilon}(b') = (1 - \varepsilon) \sum_s q_s \delta_s + \frac{\varepsilon}{|A|} \sum_{a \in A} d(a).$$

The following statements follow in a few lines, and the author should state and prove them:

- (i) A structural tie ($\delta = 0$) survives perturbation for all small $\varepsilon > 0$ if and only if $\sum_a d(a) = 0$, that is, if and only if the pair is also tied at the reference stage. Duplicates are the leading case.
- (ii) Belief-driven robust sensitivity arising from strict sign changes in δ survives for small ε .
- (iii) In the second-price auction with $T \subseteq A$, every gap bid has $d \geq 0$ with $d(b) > 0$. Hence every type is simple with $F_p(t) = \{t\}$ for every $\varepsilon > 0$ and on *every* grid containing T .

Point (iii) overturns the uniqueness of alignment as a repair. The author must then choose between two options. One is to drop the structural/belief-driven framing for the second-price auction. The other is to present it explicitly as a statement about pure level-1 conjectures, with (i) identifying the truly robust structural ties. Either choice is acceptable, but the current framing is not.

C. Position the paper in the literature (about 1.5 weeks). Write a focused related-literature section organised along the lines of Table 1. Add a short subsection, “What depth two and centredness add”, that works through the rationalizability comparison for Example 1 described in Section 6. Computing the interim correlated rationalizable sets for the running first-price example is a finite, mechanical exercise. Credit Khintchine for Lemma 1, and present $F_p(t)$ as an E-admissible set.

D. Rewrite the introduction and abstract (about 1 week). The target is an introduction of at most three pages, with one running example (Example 1 plus its cap) and an accurate statement of the assumptions and the question answered. Remove the Wilson framing, or replace it with a single accurate sentence. Replace “derived from an axiom” with an honest statement of the level-0 assumption. Move the defensive notes and the tie anatomy (Remarks 3–5, Example 5, footnote 7) to an appendix.

Recommended

E. A short computational section on predictability and revenue (about 2 weeks). Theorem 1 reduces predictability to finitely many linear sign checks, so computing $P(p)$ is cheap. The author should compute $P(p)$ and expected revenue at the uniform benchmark for first-price and second-price auctions. A suitable set of cases is $|T| \in \{3, 5, 7\}$ with equally spaced perceived values, over all two- and three-bid menus and over uniform grids of increasing fineness.

One or two figures would suffice: predictability against grid fineness, and the predictability–revenue frontier across caps. This substantiates the “trade-off frontier” that Section 7 currently only asserts. It requires no new theory, and it would turn the design discussion from anecdote into evidence.

F. Testable implications (a few days). Add a short paragraph stating falsifiable, qualitative implications. For example: in first-price auctions, bid dispersion should be larger for interior perceived types than for extreme ones; and it should fall when the menu is capped outside the dead zone. The author should say which existing laboratory designs could speak to these implications, without undertaking new empirical work.

Optional, if time remains

G. Extend the dead-zone characterisation (Corollaries 4–5) from two-bid menus to equally spaced three-bid menus. This should be attempted only if the computations in item E suggest a clean pattern. A conjecture supported by computation is an acceptable outcome.

Suggested timetable

Weeks	Tasks
1	A: reframing lemma, reorganisation, footnote 6 and minor corrections
2–3	B: perturbed-anchor proposition; rewrite the second-price section accordingly
4	C: literature section and rationalizability comparison
5–6	E: computational section; F: testable implications
7	D: new introduction and abstract; streamline the main text; move material to appendices
8	Buffer; G if feasible; full proofreading and consistency check of cross-references

10 Summary of required changes

1. State that robust simplicity is independent of the centred class, and reorganise the results into class-free and class-dependent groups.
2. Confront the fragility of structural ties under perturbed anchors, and revise the second-price design conclusions accordingly.
3. Correct footnote 6, and fix the minor inaccuracies in Section 4.3.
4. Position the paper against rationalizability, robust predictions, level- k , analogy-based reasoning, E-admissibility and Khintchine’s theorem.
5. Bring the abstract, introduction and verbal interpretations into line with the formal results. In particular, do not claim that belief-driven ties are “exceptional”, and do not claim that sensitivity predicts bids in the gaps.
6. Substantially shorten and clarify the exposition.

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Table 1: Departures from standard theory and closest literature

Departure	Closest works	Relation to this paper
No common prior; robust predictions	[9, 27, 1, 2]	Canonical answers to the opening question. <i>Not cited.</i>
Level- k with uniform level-0	[20, 24, 6, 8, 7]	Anchors are level-1; the case $q = p$ is level-2. Crawford–Iriberri study exactly this in auctions.
Mixture of lower levels	[5]	Full-support conjectures would remove second-price structural ties.
Beliefs tilted toward own type	[12, 23]	Projected beliefs are centred, but are used in equilibrium.
Coarse perception of opponents	[14, 15, 11]	Jehiel–Koessler (coarse types in incomplete-information games) is closest. <i>Not cited.</i>
Finite depth and higher-order beliefs	[25, 13]	Robustness of finite-depth reasoning. <i>Not cited.</i>
Sets of priors; set-valued choice	[17, 26, 3, 19]	$F_p(t)$ is Levi’s E-admissible set. <i>Not cited.</i>
Misspecified-equilibrium closure	[10]	The “first closure” of Section 7. <i>Not cited.</i>
Simplicity and reasoning depth	[18, 4, 22]	Pycia–Troyan tie simplicity to foresight depth. <i>Not cited.</i>
Lemma 1	[16, 21]	Discrete Khintchine representation. <i>Not credited.</i>